

Saketh Elumalai

408-219-9754 | elu.saketh@gmail.com | [linkedin.com/in/saketh-elumalai/](https://www.linkedin.com/in/saketh-elumalai/) | github.com/hype2056

EDUCATION

Irvington High School

Fremont, CA

Jun. 2028

Ohlone College

Materials Engineering

Fremont, CA

Aug. 2025

PROJECTS

Compify | *Python, Flask, React, TypeScript, Vite, Docker*

Sep. 2025 – Feb. 2026

- Built an AI-powered competition math tutoring web app to submit math problems via text or image upload and receive structured solutions.
- Integrated **agentic RAG** with structured output and thinking mode to retrieve AOPS competition problems
- Implemented **multimodal OCR and solution verification** by passing base64-encoded images directly to vision model for high-accuracy math notation transcription
- Designed an interactive practice system with a PracticeModal component deployed to production on Vercel with **localStorage-based API** key persistence for seamless user sessions.

Wildfire Tracker | *Next.js, NASA FIRMS API, Leaflet.js, PostgreSQL*

Feb. 2026

- Built a **real-time wildfire map for California** using NASA FIRMS satellite data with 24-hour refresh
- Implemented brightness filtering to **surface high-intensity fire detections** from live sensor data
- Deployed to Vercel with **environment-secured API keys** and a public GitHub repository

Neural Network from Scratch | *Python, NumPy, Matplotlib*

Mar. – Apr. 2026

- Implemented a feedforward neural network in pure NumPy with no ML frameworks - **forward pass, backpropagation, and gradient descent** from scratch
- Trained on 41,000 MNIST handwritten digit images, achieving **89% accuracy over 500 iterations**
- **Built ReLU and softmax activation functions manually** to understand how modern networks actually work

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, MatLab

Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, Eclipse